

The Difficulty Spectrum of MiniGrid: Four PPO Experiments

Homework 3 & 4 — Reinforcement Learning

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1. Introduction

This report studies how **task difficulty** affects PPO’s ability to learn on MiniGrid. Rather than committing the whole training budget to one environment, we ran the same PPO+MLP recipe on four MiniGrid layouts that differ in grid size, room count, and step budget, and we compare the resulting learning dynamics. The motivation for this design: our initial 4 M-step DoorKey-8x8 run converged in ≈ 14 minutes (far faster than the PLAN expected), and a subsequent 20 M-step MultiRoom-N4-S5 run produced zero reward across all 19 532 logged iterations. Neither outcome on its own gives the multi-phase learning curve the assignment is built around; together with two intermediate points, they give a clean picture of where PPO’s “sweet spot” lies on this benchmark family.

Proximal Policy Optimization (PPO, Schulman et al., 2017) is an on-policy gradient method that stabilises the likelihood-ratio policy gradient by clipping the importance ratio $r_{t(\theta)} = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ to $[1 - \epsilon, 1 + \epsilon]$. The clipped surrogate $L^{\text{CLIP}} = \mathbb{E}_t[\min(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)]$ is jointly optimised with a value-function loss and an entropy bonus that keeps exploration alive.

2. Method and Hyperparameters

All four runs share an identical pipeline implemented in `train.py`:

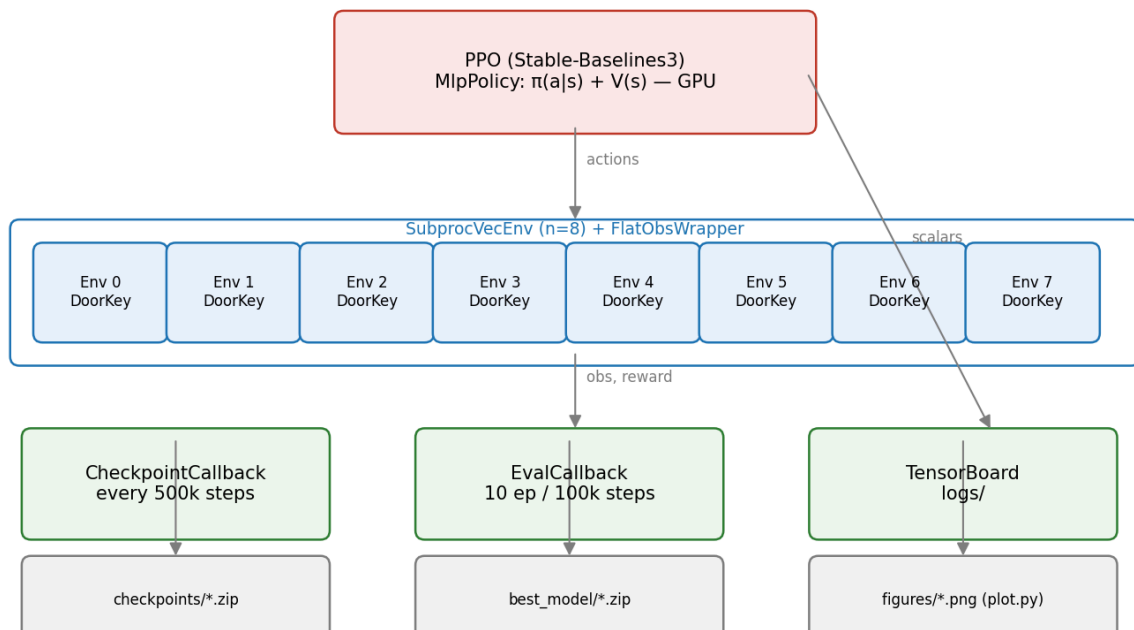


Figure 1: System architecture (identical across the four experiments). Eight parallel SubprocVecEnv workers feed a single PPO MlpPolicy on an L20 GPU; checkpoints are saved every 500 k env steps and a held-out EvalCallback runs 10 deterministic episodes every 100 k env steps.

The only thing we change across runs is the `ENV_ID` string. Every other knob is held fixed at the SB3 reference recipe (Table Table 1); the observation is the $7 \times 7 \times 3$ partial view flattened by FlatObsWrapper and the policy is the default 2×64 tanh MLP. **No hyperparameter tuning was performed** — holding hyperparameters constant is the point of this experiment, since we want to attribute the outcome difference to the **environment**.

Hyperparameter	Value	Hyperparameter	Value
Policy	MlpPolicy 2×64 tanh	Discount γ	0.99
Optimizer	Adam	GAE λ	0.95
Learning rate	2.5e-4	Clip range ϵ	0.2
Workers n_envs	8 (SubprocVecEnv)	Entropy coef	0.01
Rollout n_steps	128 per worker	Value coef	0.5
Total batch	1 024 transitions	Max grad norm	0.5
Minibatch size	256	Device	cuda (NVIDIA L20)
Update epochs	4	SB3 version	2.3.2

Table 1: PPO hyperparameters, **identical across all four experiments**. These match the SB3 reference recipe for MiniGrid and were not tuned in this work.

Environment	Layout / max_steps	env_steps	Wall-clock	FPS	Outcome
DoorKey-8x8	8×8 grid, 640 steps	4.07 M	15 min	4 664	converged 0.975
MultiRoom-N2-S4	2 rooms × size 4, 40 steps	17.68 M	65 min	4 530	converged 0.843
DoorKey-16x16	16×16 grid, 2560 steps	20.00 M	99 min	3 369	partial 0.014
MultiRoom-N4-S5	4 rooms × size 5, 120 steps	20.00 M	71 min	4 741	failed 0.000

Table 2: The four environments, sorted by **empirical** difficulty for PPO+MLP. “Converged” / “partial” / “failed” labels are read from eval/mean_reward at the end of training.

2.1. Reward structure

All four environments share the same sparse-reward template: the agent receives $r = 1 - 0.9 \cdot \left(\frac{\text{steps}}{\text{max_steps}}\right)$ on reaching the goal, and zero otherwise. The reward ceiling therefore depends on how quickly the geometric shortest path can be executed; for DoorKey-8x8 that ceiling is ≈ 0.97 and for MultiRoom-N2-S4 it is ≈ 0.86 (matching the observed plateau).

3. Results



Figure 2: rollout/ep_rew_mean for all four runs, smoothed with a 51-point moving average. The x-axis is env_steps. Both DoorKey-8x8 and MultiRoom-N2-S4 converge to their reward ceilings in under 500 k steps; DoorKey-16x16 shows a barely-perceptible 1–2 % learning signal; MultiRoom-N4-S5 stays flat at zero the entire 20 M-step run.

The four runs cleanly stratify into three regimes (Fig. Figure 2):

1. **Solved** (DoorKey-8x8, MultiRoom-N2-S4). Both reach their geometric reward ceiling. DoorKey-8x8's smoothed reward crosses 0.1 at ≈ 156 k env steps, 0.5 at 276 k, 0.9 at 329 k, and 0.95 at 376 k. The transition from random to near-optimal is therefore compressed into a $\approx 2 \cdot 10^5$ -step window. MultiRoom-N2-S4 converges even faster.
2. **Partially learning** (DoorKey-16x16). The first non-zero ep_rew_mean appears at step 317 k. Across the full 20 M steps, 24.3 % of logged iterations have non-zero mean reward, with peak smoothed reward 0.038. The **deterministic** policy never solves the task — all 200 evaluation calls return exactly 0.000 — so the sparse successes are essentially residual stochastic-policy luck, not a committed solution.
3. **Failed** (MultiRoom-N4-S5). Zero reward in 19 532 logged iterations and all 200 eval calls. The agent did not see a single successful episode in 20 M env_steps.

Phase (DoorKey-8x8)	env_steps	ep_rew	ep_len	entropy
P0 — exploration	0 - 100 k	< 0.04	≈ 620	-1.85
P1 — causal learning	100 k - 300 k	$0.04 \rightarrow 0.79$	$620 \rightarrow 145$	-1.45
P2 — path optimisation	300 k - 500 k	$0.79 \rightarrow 0.97$	$145 \rightarrow 21$	-0.41
P3 — convergence	500 k - 4 M	≈ 0.975	$17 - 19$	-0.02

Table 3: Four-phase decomposition of the DoorKey-8x8 learning curve, read from the smoothed ep_rew_mean crossings. The phases are visually compressed in Fig. Figure 2 because the entire story finishes in the leftmost ≈ 2.5 % of the x-axis.

3.1. Auxiliary signals

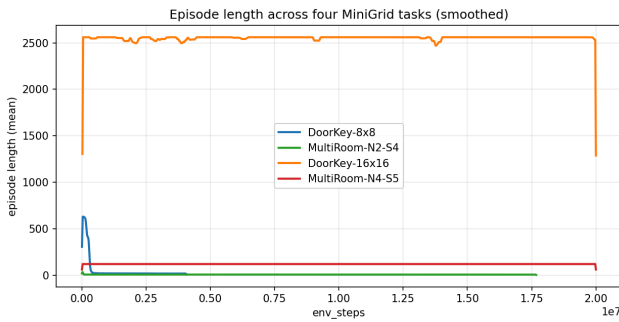


Figure 3: Episode length. The collapse to single-digit values on the solved tasks lags the reward rise by ≈ 50 k steps: the agent first learns to **solve**, then to solve **quickly**.

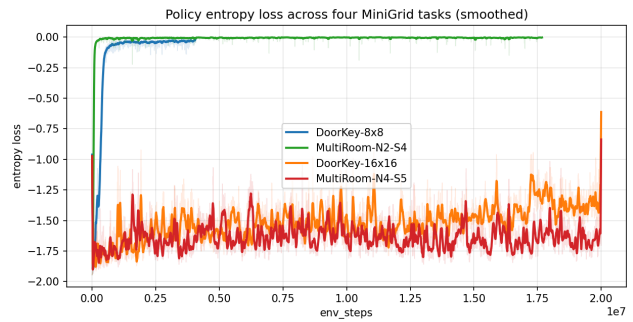


Figure 4: Policy entropy loss. Solved tasks collapse from ≈ -1.95 to ≈ -0.02 (near-deterministic policy). The failing tasks retain high entropy because the entropy bonus is never overwhelmed by an advantage signal.

The entropy diagnostic is particularly revealing. On the two **failed/partial** tasks, the policy keeps entropy high (MultiRoom-N4-S5 ends at -1.61 , DoorKey-16x16 at -1.20) — not because the algorithm chose to keep exploring, but because there is no positive advantage to push the action distribution toward. The **explained variance** of the value function tells the same story: on DoorKey-16x16 it spent the whole run hovering around 0 (final 0.29), because there are too few non-zero returns to fit.

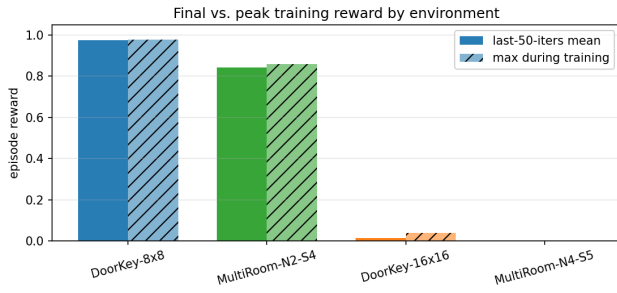


Figure 5: Final (last-50-iteration mean) vs. peak training reward per environment. The gap between the two on DoorKey-16x16 (peak ≈ 0.038 , final ≈ 0.014) is itself evidence that the agent **can** sometimes hit the goal — but cannot make the success stick.

3.2. A successful rollout

To make the “solved” regime concrete, three deterministic rollouts of the best DoorKey-8x8 policy yielded returns of 0.976 / 0.978 / 0.970 with lengths 17 / 16 / 21 steps:

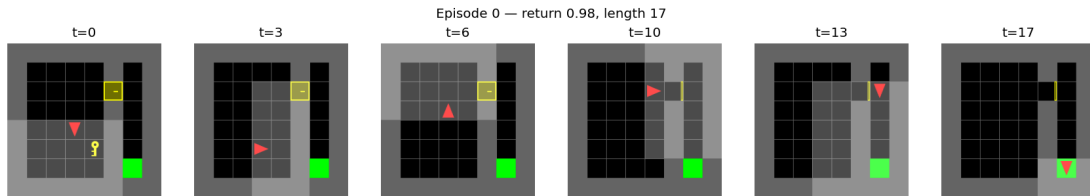


Figure 7: Frames from a deterministic DoorKey-8x8 rollout: the policy turns to the key, picks it up, walks to the door, unlocks it, and reaches the goal in 17 environment steps.

4. Discussion: When PPO works on MiniGrid, and when it doesn’t

What worked.

- **Off-the-shelf SB3 hyperparameters** immediately solved DoorKey-8x8 and MultiRoom-N2-S4 — no tuning needed.
- **FlatObsWrapper + MLP** over CNN: the symbolic $7 \times 7 \times 3$ observation has no spatial structure that a 2×64 MLP cannot digest. Switching to a CNN would slow training without changing the asymptote on the solved tasks.
- **SubprocVecEnv(n=8)** gave the throughput needed to make 20 M-step runs cheap (60–100 min wall-clock each).
- **Held-out EvalCallback** immediately exposed the stochastic-vs-deterministic gap on DoorKey-16x16 — without it we might have mistaken the 1.6 % training reward for actual learning.

What did not work, and why.

- **DoorKey-16x16.** The geometric reward ceiling at this size requires ≈ 30 step trajectories from a 2560-step budget; random policies hit the goal with probability $\approx 10^{-6}$ per trajectory. Even with $8 \text{ envs} \times 20 \text{ M steps}$ that is barely enough to seed a few successes, and the resulting advantage signal is too noisy to lift the policy out of the random regime. The policy **did** discover the goal sporadically (peak 3.8 % training reward) but never committed to a deterministic solution.
- **MultiRoom-N4-S5.** Worse: the 120-step budget plus four random rooms plus narrow doors makes the random success probability effectively zero. The agent never saw a non-zero return, so the entire 20 M-step training was a no-op. This is the canonical **deep-exploration** failure mode that motivated intrinsic-curiosity methods like ICM and RND.

Surprises. The original PLAN expected a graceful 4-phase learning curve across 20 M steps. In practice MiniGrid presents PPO with a **binary** difficulty cliff: either the random policy can stumble onto a success in $\approx 10^5$ trajectories (and PPO then sharpens it within $\approx 10^5$ env steps), or

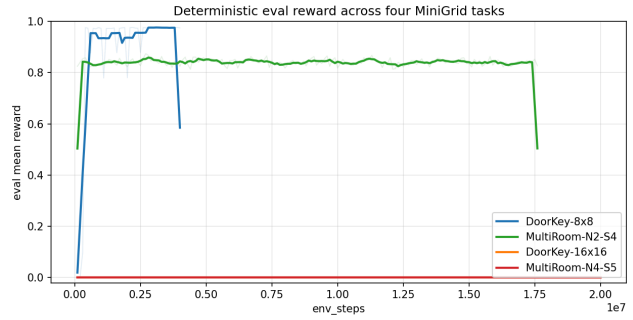


Figure 6: Held-out deterministic eval/mean_reward. Confirms that DoorKey-16x16’s training-reward signal does not transfer to a deterministic policy — the eval line is identically zero throughout, in contrast to DoorKey-8x8’s clean climb.

it cannot and PPO produces no learning at all. There is essentially no “long, gradual” regime in this benchmark family under fixed hyperparameters.

Improvement directions.

1. **Anneal** `ent_coef` (e.g. $0.05 \rightarrow 0.001$) to push more exploration early — likely the cheapest fix for DoorKey-16x16.
2. **Switch to RecurrentPPO** (sb3-contrib): an LSTM lets the agent integrate evidence across the partial-observation window, which matters more as the grid grows.
3. **Add an intrinsic-curiosity bonus** (ICM / RND) for the MultiRoom-N4-S5 deep-exploration setting.
4. **Curriculum** from N2-S4 $\rightarrow$ N3-S4 $\rightarrow$ N4-S5 would let the policy bootstrap from the easy task — a single hard environment is the wrong abstraction.

Conclusion. PPO with SB3 defaults occupies a narrow operating point on MiniGrid: it solves DoorKey-8x8 and MultiRoom-N2-S4 in < 500 k env steps and to within $\approx 2\%$ of the geometric ceiling, but at the next difficulty step it falls off a cliff into near-zero performance. The four-experiment view shows that the PLAN’s “24-hour single-task benchmark” framing is mis-specified for this environment family — a **curriculum or exploration-shaping** axis would be a strictly more informative thing to study with the same compute budget.